MACHINE BIAS IN CRIME PREDICTION

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1

PROPUBLICA STUDY

COMPAS, BY NORTHPOINTE

What is it?

A computer program that predicts the likelihood of a person to commit a future crime

What is it for?

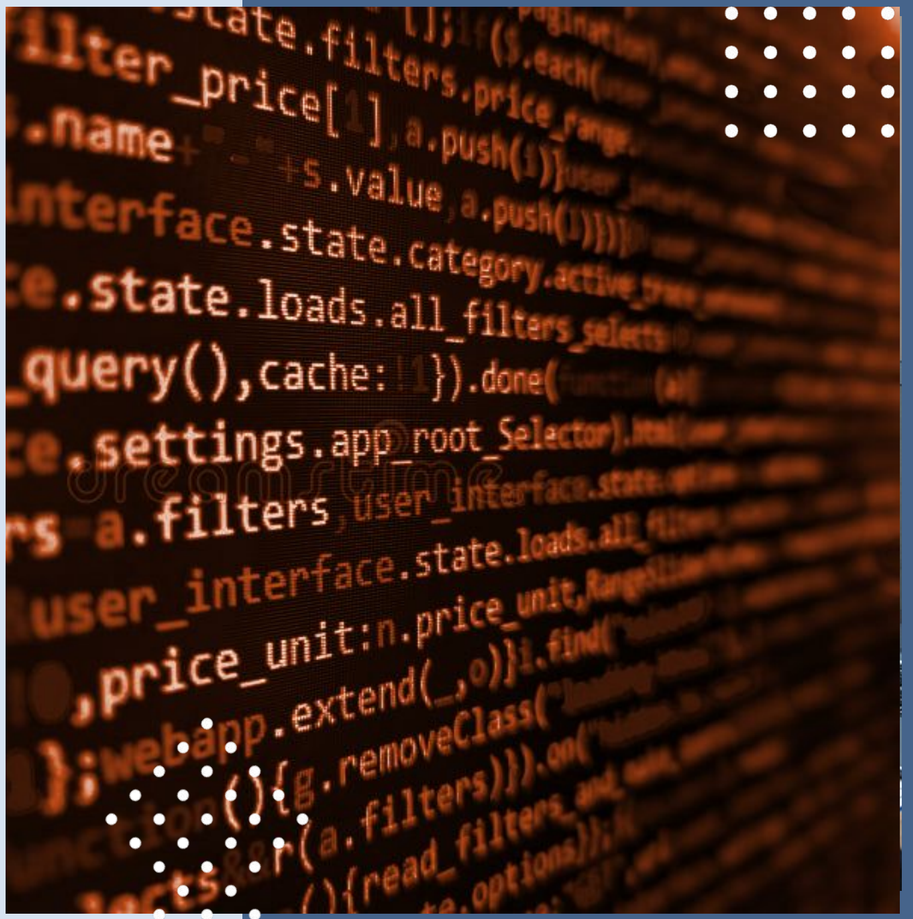
Used in courtrooms, to inform parole

Why do we care?

No prior studies on bias within the scores

Why study it?

The program “May exacerbate unwarranted and unjust disparities”



SPECIFICS

- **ProPublica study defines recidivism** *as a criminal offense that resulted in a jail booking and took place after the crime for which the person was COMPAS scored.*

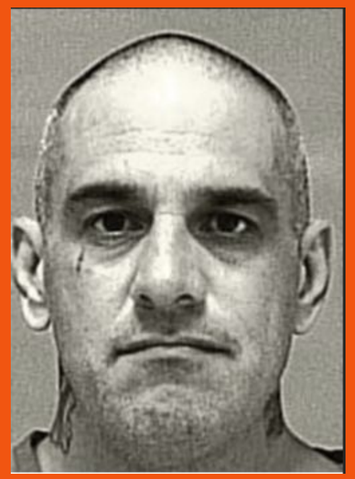
- **Recidivism risk scores (1-10 scale):** (1) All and (2) Violent risk scores calculated using questionnaire, answered by defendants.
  - ‘Was one of your parents ever sent to jail/prison?’
  - ‘How often did you get in fights while at school?’
  - ‘Does a hungry person have a right to steal?’

- **Observed Recidivism:** records whether a person had actually committed another crime within **two years** of being released

Race is NOT a question

SAME CRIME, DIFFERENT OUTCOMES

RISK SCORE: 3



Vernon Prater

Shoplifted **$80** of tools
Priors: **armed robbery,** 5 years in prison

Later stole **1000s of dollars** worth of electronics

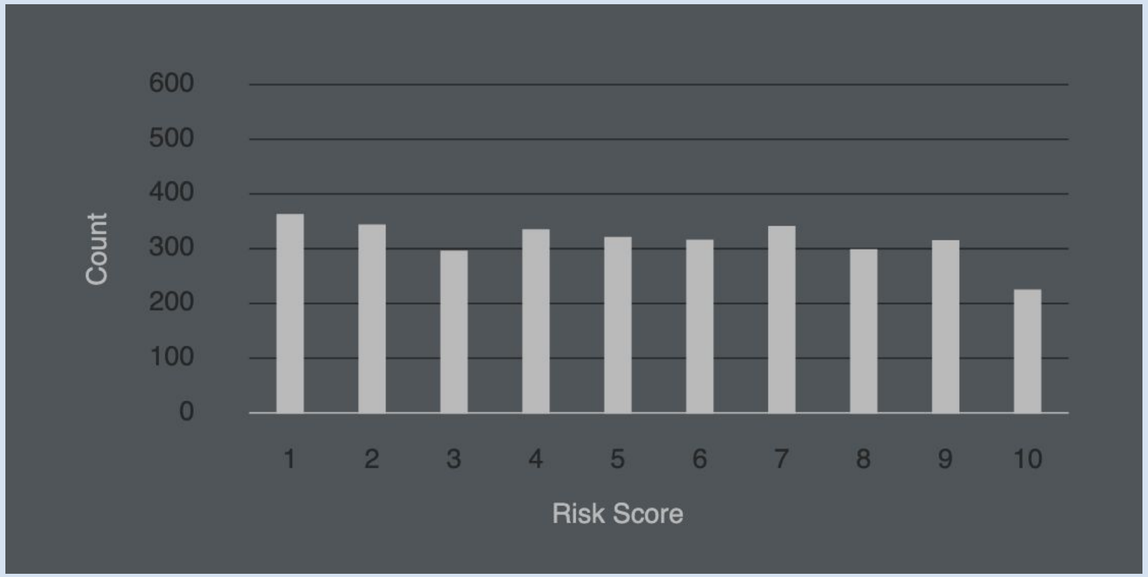
RISK SCORE: 8

Brisha Borden

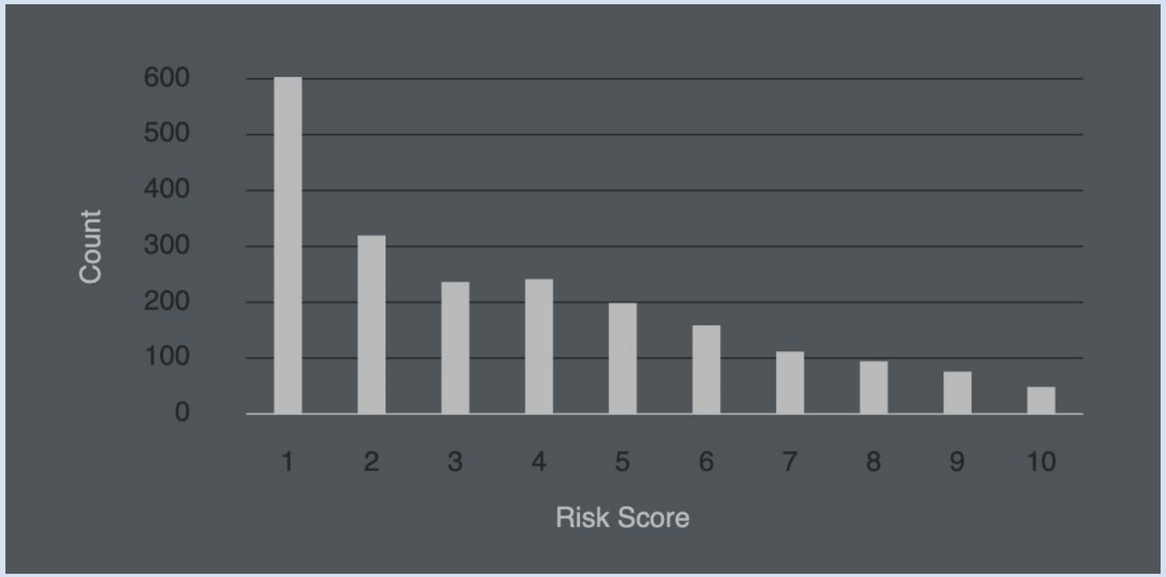
Stole a kid's **bike** and **scooter**
A record of juvenile misdemeanors

Did NOT reoffend

RISK SCORE COMPARISONS



Black Defendants' Risk Scores



White Defendants' Risk Scores

Larson et al. 2014, How We Analyzed the COMPAS Recidivism Algorithm

“A GUY WHO HAS MOLESTED A SMALL CHILD EVERY DAY FOR A YEAR COULD STILL COME OUT AS A LOW RISK BECAUSE HE PROBABLY HAS A JOB,” BOESSENECKER SAID. “MEANWHILE, A DRUNK GUY WILL LOOK HIGH RISK BECAUSE HE’S HOMELESS. THESE RISK FACTORS DON’T TELL YOU WHETHER THE GUY OUGHT TO GO TO PRISON OR NOT; THE RISK FACTORS TELL YOU MORE ABOUT WHAT THE PROBATION CONDITIONS OUGHT TO BE.”

TAKEAWAYS

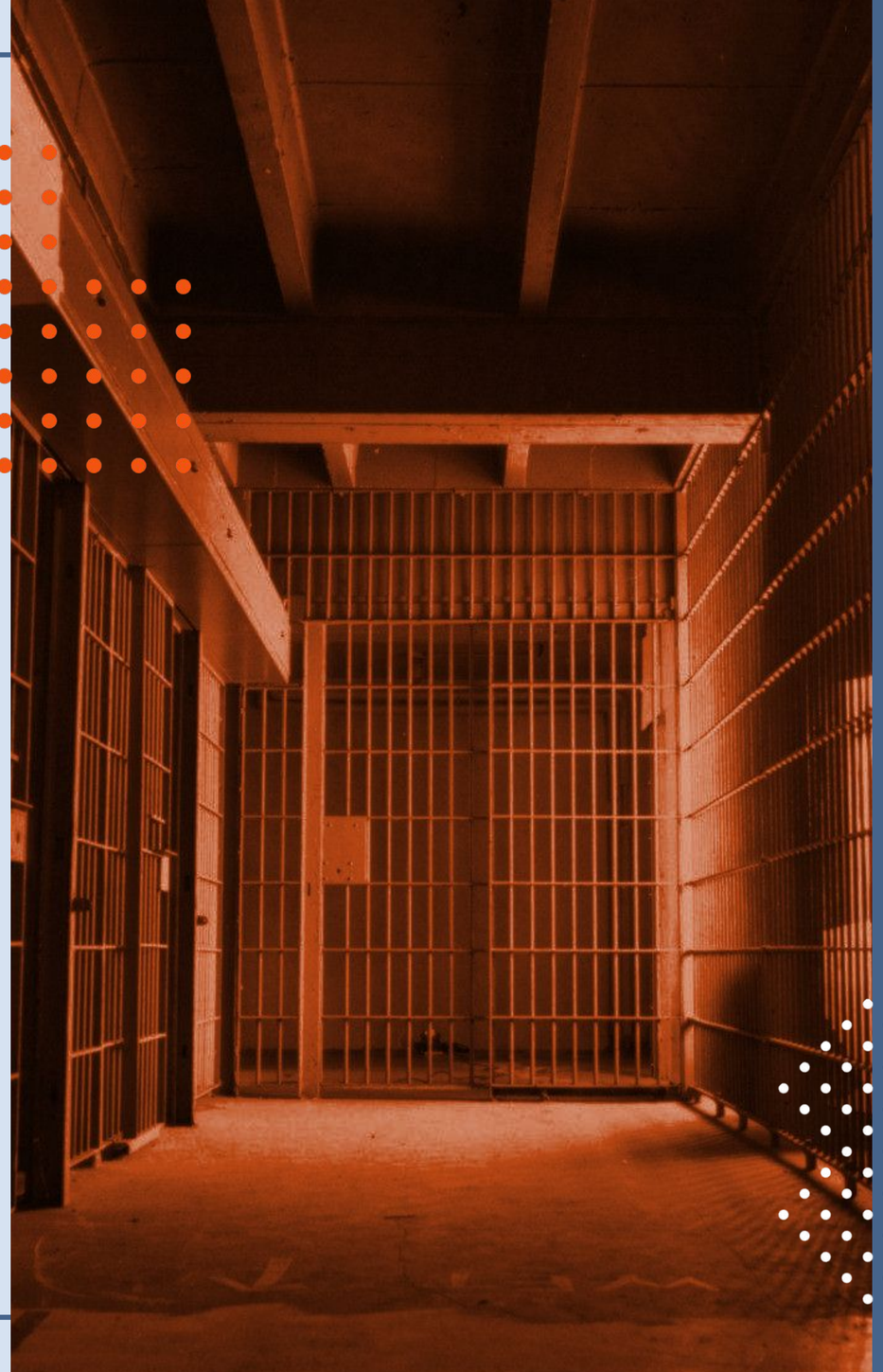
1. Black defendants were 77% more likely to be labelled as higher risk of committing a violent crime in the future

2. Black defendants were 45% more likely to be likely to be labelled as higher risk of committing any crime in the future

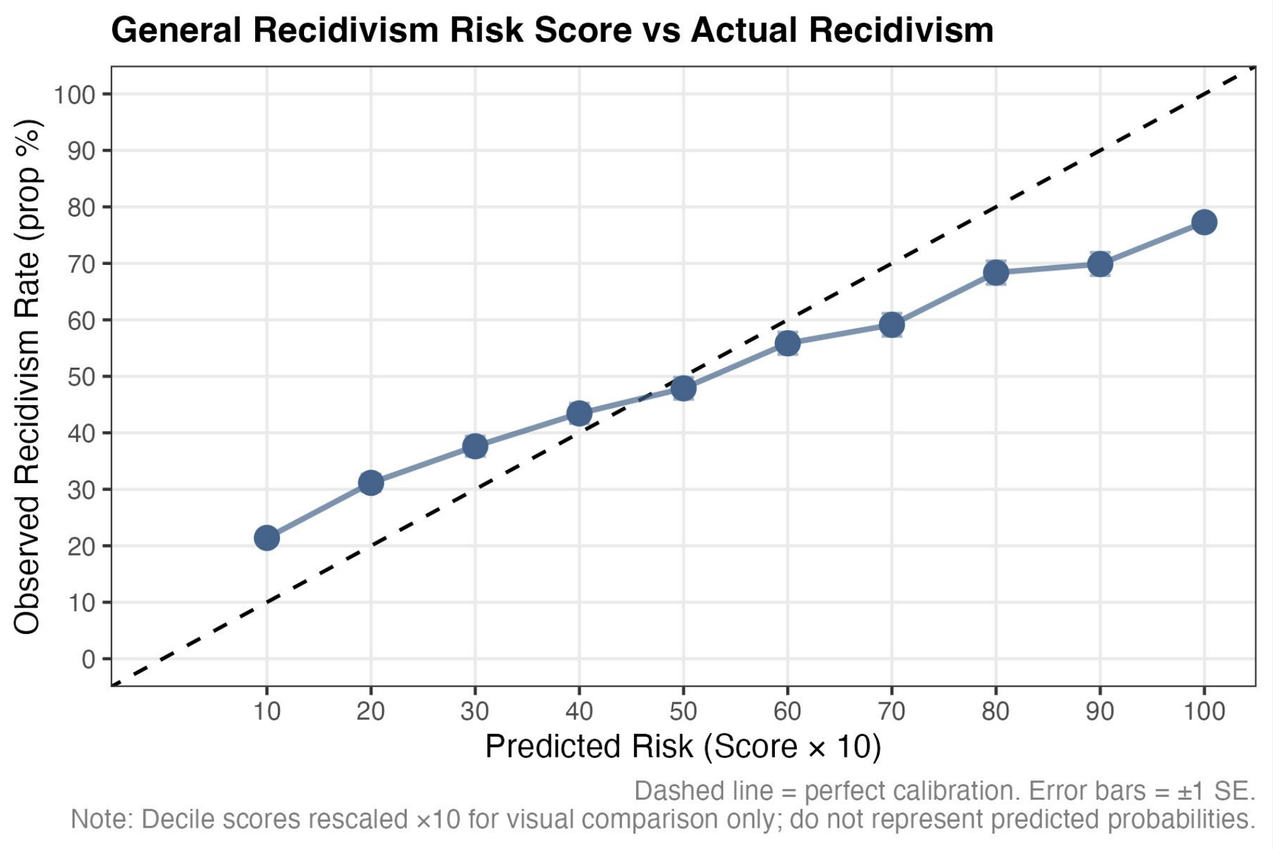
3. **Risk scores significantly influence how long people will stay in prison and what your parole looks like**

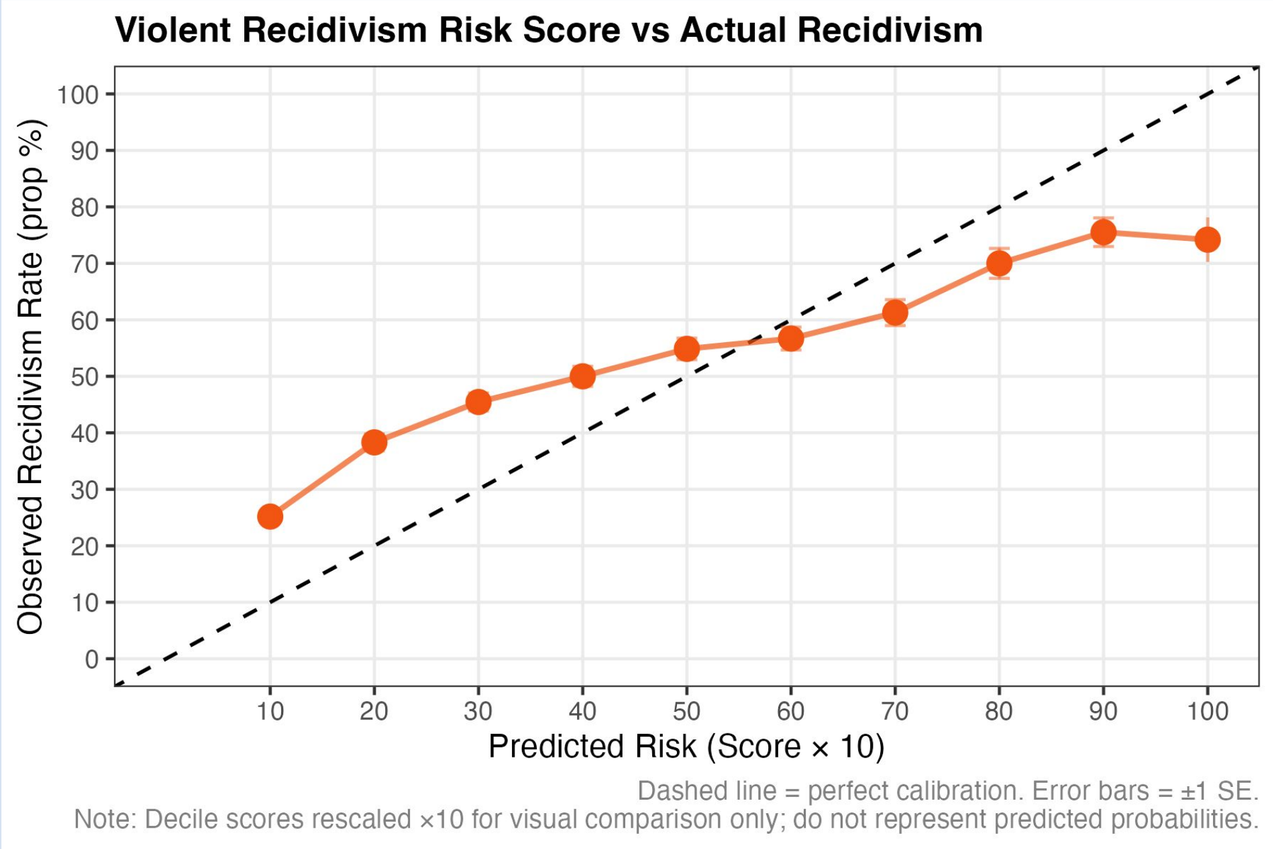
2

DATA EXPLORATION



HOW DO RISK SCORES CORRELATE WITH ACTUAL RECIDIVISM?





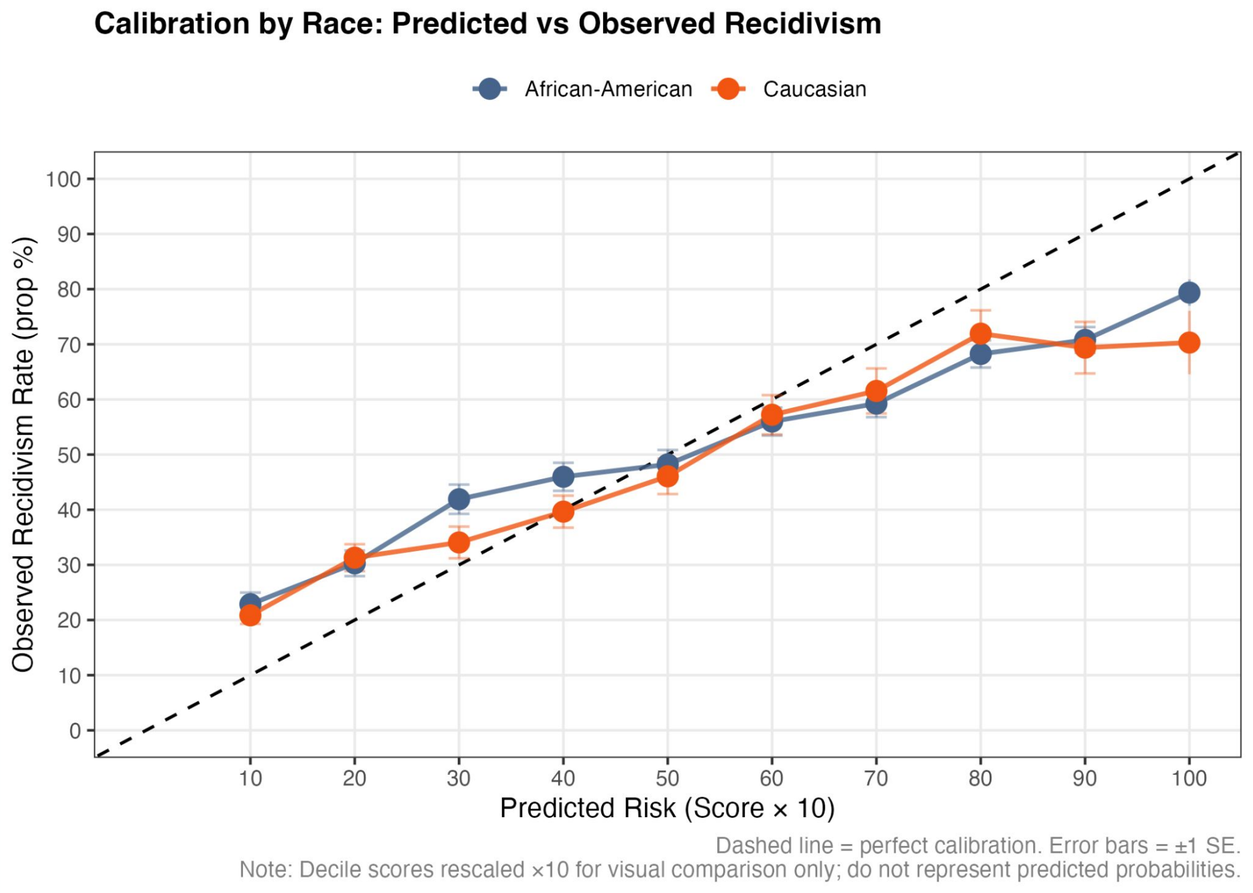
Interpretation:

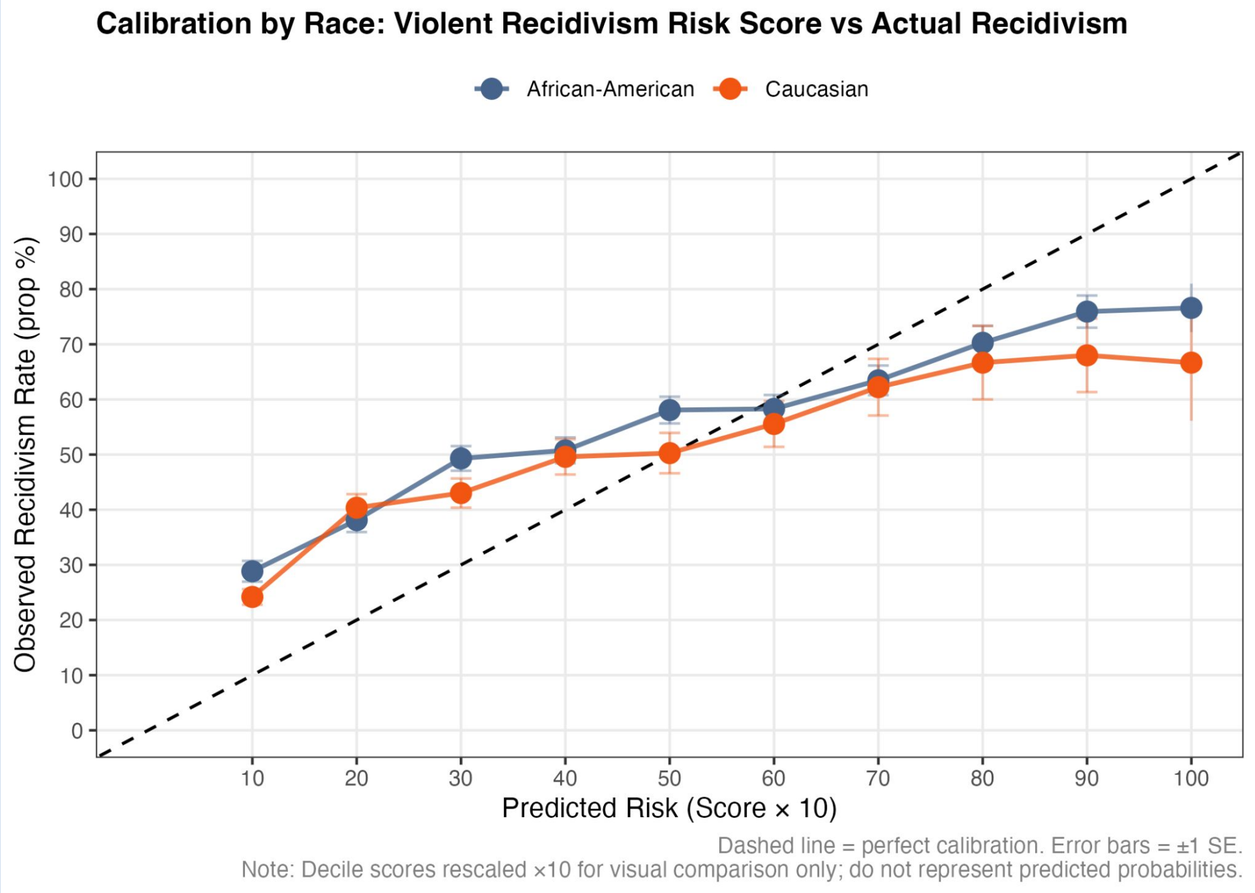
What percentage of people with a predicted score of [1-10] have actually reoffended within two years of being released?

Answers:

Does COMPAS actually predict recidivism accurately, and where does it break down?

HOW DOES ACCURACY CHANGE WITH RACE INVOLVED?



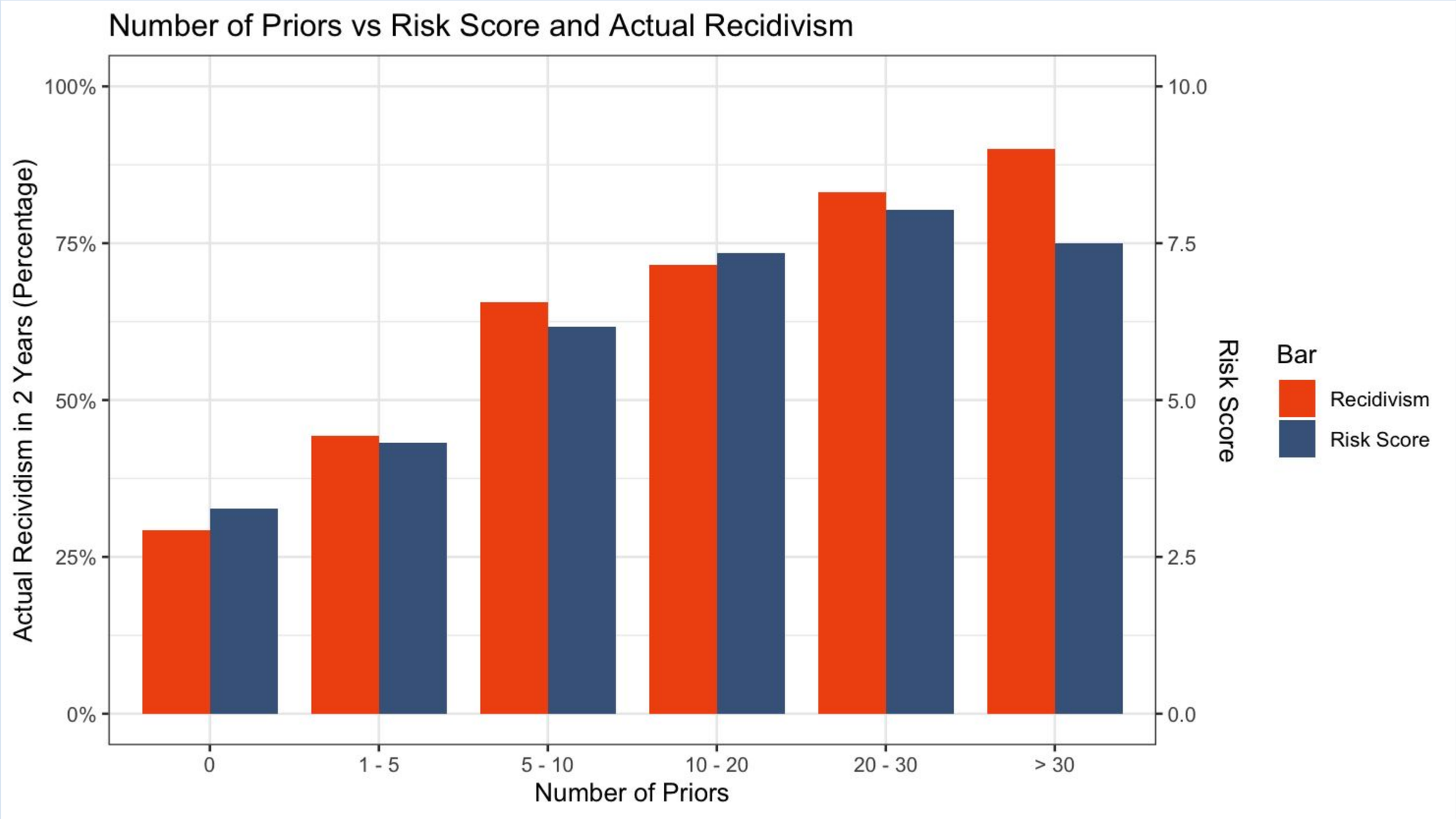


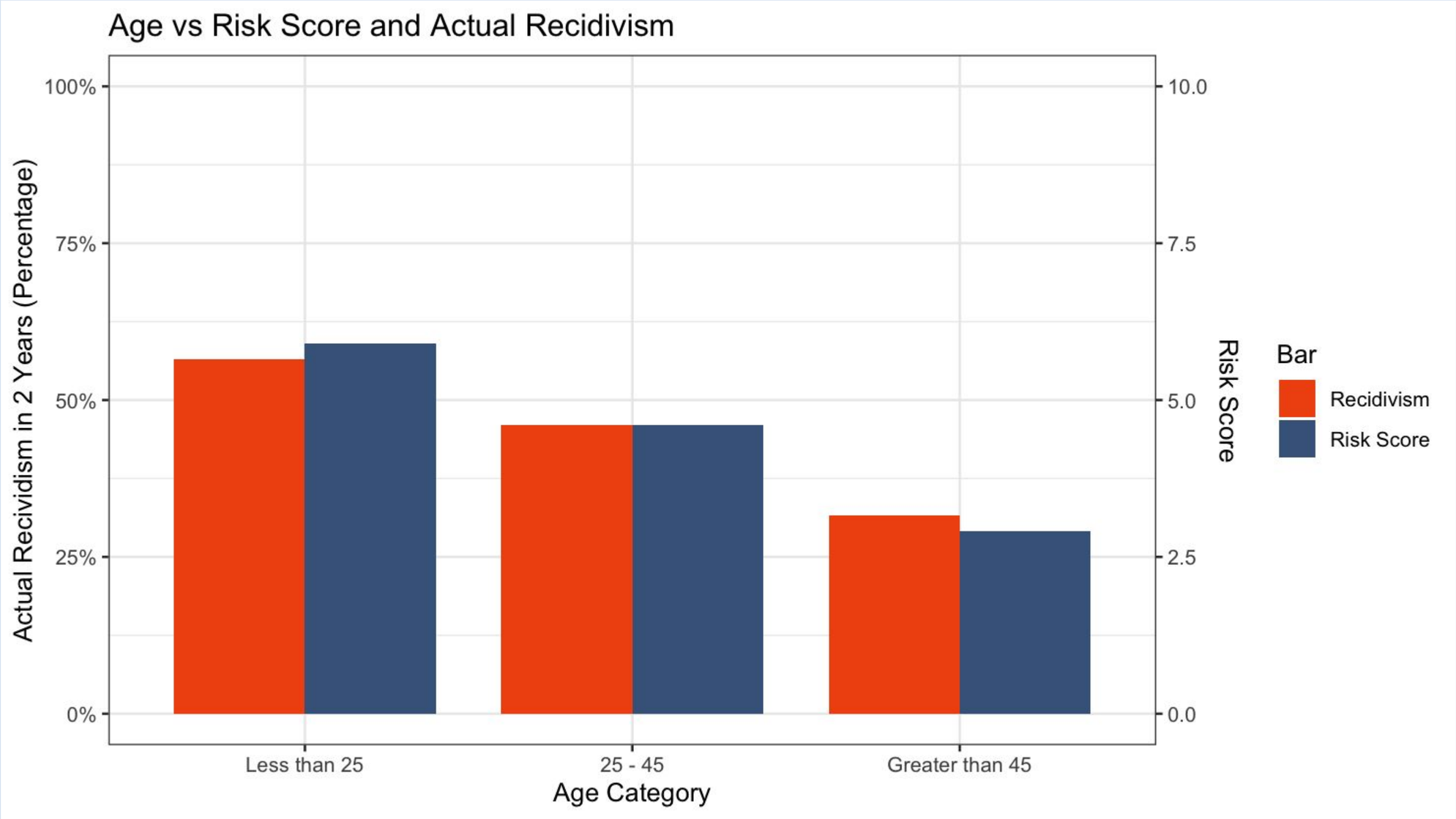
Interpretation:

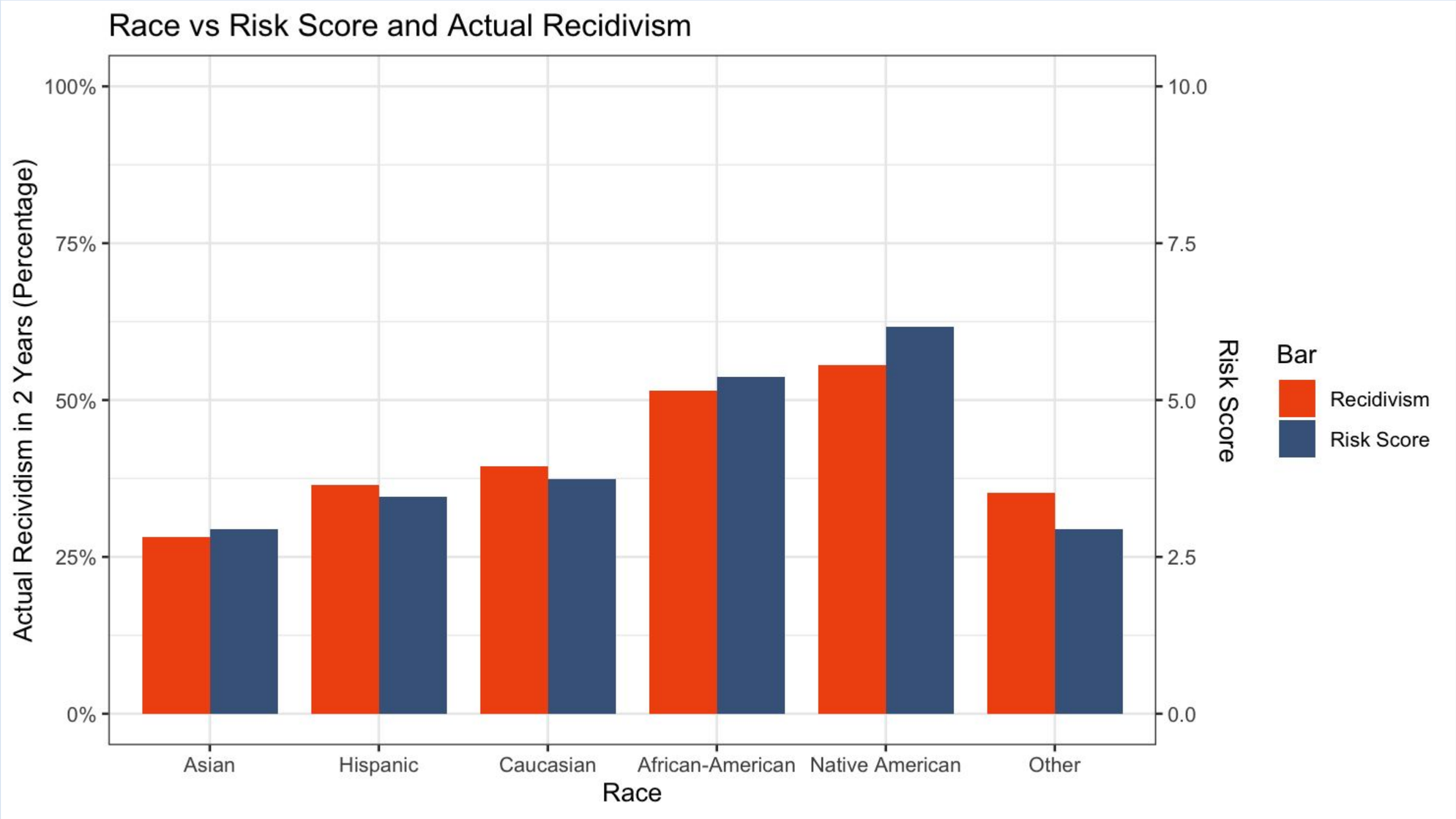
Among Black and White defendants assigned the same risk score [1-10], what percentage of each group actually reoffended within two years. Does this relationship hold equally across both general and violent recidivism scores?

Answers:

Does COMPAS predict recidivism equally well for Black defendants as it does for White defendants at the same score level?


Number of Priors vs Risk Score and Actual Recidivism
Actual Recividism in 2 Years (Percentage)
100%
75%
50%
25%
0%
10.0
7.5
5.0
2.5
0.0
Risk Score
0
1 - 5
5 - 10
10 - 20
20 - 30
> 30
Number of Priors
Bar
Recidivism
Risk Score


Age vs Risk Score and Actual Recidivism
Actual Recividism in 2 Years (Percentage)
100%
75%
50%
25%
0%
10.0
7.5
5.0
2.5
0.0
Risk Score
Less than 25
25 - 45
Greater than 45
Age Category
Bar
Recidivism
Risk Score


Race vs Risk Score and Actual Recidivism
Actual Recividism in 2 Years (Percentage)
100%
75%
50%
25%
0%
10.0
7.5
5.0
2.5
0.0
Risk Score
Asian
Hispanic
Caucasian
African-American
Native American
Other
Race
Bar
Recidivism
Risk Score

WHICH CHARGE TYPES ARE MOST LIKELY TO BE OVER-LABELED ?

- Compare the medium and high-risk rate(general recidivism score) and the two-year recidivism rate within each charge type
- Construct a gap measure:
  High-risk rate - Actual recidivism rate
- A positive gap suggests over-labeling into the high risk

MAIN FINDINGS

Largest positive gaps:

- arrest case no charge: +0.070
- Grand Theft (Motor Vehicle): +0.054
- Pos Cannabis W/Intent Sel/Del: +0.043
- Grand Theft in the 3rd Degree: +0.029

Negative gaps:

- Battery: −0.070
- Felony Battery (Dom Strang): −0.141
- Driving Under The Influence: −0.190
- Felony Driving While Lic Suspd: −0.194

WHAT FACTORS MOST STRONGLY PREDICT A HIGH-RISK LABEL?

METHOD

Random Forest classifier

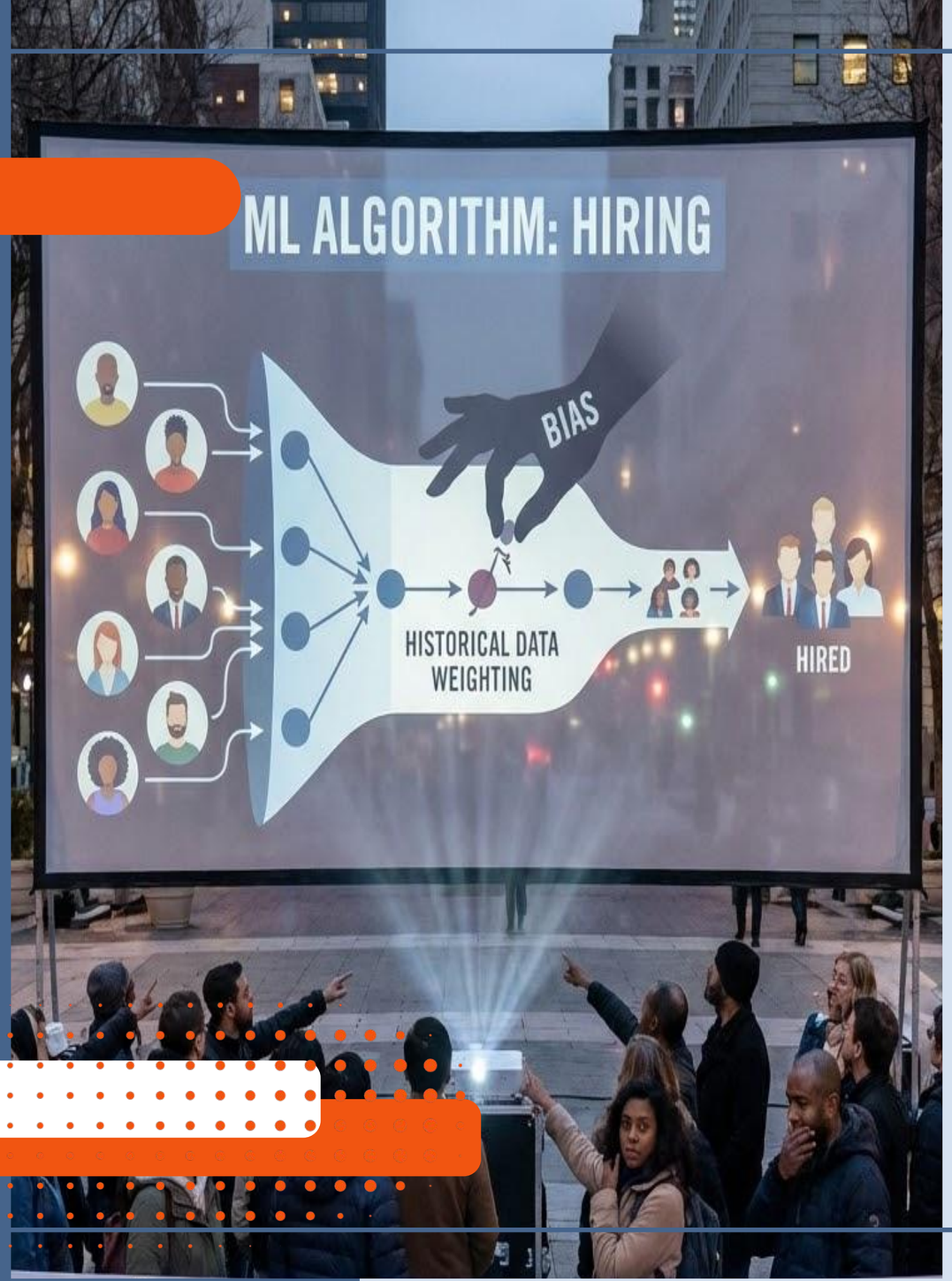
TOP FEATURE IMPORTANCE

priors_count = 0.329
age = 0.289
two_year_recid = 0.148
race = 0.050
charge degree, sex, and specific charges follows

PERFORMANCE

Accuracy: 0.754

AUC: 0.846



How does Bias creep into Crime Prediction Models?

BIAS CAN CREEP IN ANY STAGE

1. Training Data - Historical policing patterns, Record Arrest Data etc. Feature Selection - neighbourhood, employment.

2. Model Training on the biased data.
The Risk Prediction Model estimates Risk Scores.

3. Judicial decision.
Social Outcome is reinforced inequality?

WHY BOTH SIDES APPEAR CORRECT

PROPUBLICA

Focus: Error Rates

Black defendants had more false positives.

COMPAS

Focus: Calibration

Same risk score implies same probability of reoffending.

3

POLICY IMPLICATIONS

BENEFITS

- Algorithms can improve consistency in decision-making by ensuring that similar cases are evaluated using the same statistical criteria.
- They may reduce arbitrary human bias by applying standardized rules rather than relying solely on individual judgment.
- Machine learning models can detect patterns in large datasets that may not be easily observable to human decision-makers.
- These systems can support better allocation of limited resources by identifying areas or individuals that may require greater attention.
- Risk assessment tools may also help identify individuals who could benefit from rehabilitation or support programs.

RISKS AND CONCERNS

- Algorithms may reproduce inequalities present in historical data.
- Judges and other decision-makers may place excessive trust in algorithmic predictions, a phenomenon known as automation bias.
- Many algorithmic risk assessment systems are proprietary, making them difficult to audit or evaluate independently.
- Incorrect predictions generated by these systems can directly influence decisions about bail, sentencing, or parole.
- When used widely, these systems may reinforce existing systemic inequalities in the criminal justice system.

4

POLICY INTERVENTIONS

POLICY RESPONSES IMPLEMENTED

- Courts in several jurisdictions treat algorithmic risk scores (such as COMPAS) as advisory tools rather than deterministic decision rules, meaning judges are encouraged not to rely on them exclusively.
- Wisconsin Supreme Court (State v. Loomis, 2016) ruled that COMPAS can be used in sentencing, but judges must be warned about its limitations and should not rely on it as the sole basis for decisions.
- Some states and counties have introduced guidelines requiring judicial discretion when using risk assessment tools in bail or sentencing decisions.
- New York City's Algorithmic Accountability Law (2018) requires city agencies to review automated decision systems and evaluate potential bias and impacts on civil rights.
- Several U.S. jurisdictions have begun implementing algorithmic auditing practices, where external researchers or oversight bodies evaluate predictive systems for disparate impacts across demographic groups.
- The NIST AI Risk Management Framework (2023) is now used by many U.S. government agencies and organizations to guide the responsible development and deployment of AI systems, including bias testing and risk monitoring.

KEY TAKEAWAYS

- Predictive models must have tactical utility, because even highly accurate algorithms are not useful if they do not meaningfully guide policing decisions or interventions.
- High-quality and representative data are essential, since models trained on biased or incomplete historical data may reproduce patterns of past policing rather than patterns of crime.
- Algorithmic predictions should not be mistaken for causal explanations, because machine learning models detect correlations rather than the underlying drivers of criminal behavior.
- Predictive systems require continuous evaluation and auditing, as deploying models without rigorous assessment can reinforce existing enforcement patterns instead of improving outcomes.
- Civil liberties and public trust must remain central considerations, because large-scale deployment of predictive technologies can amplify surveillance, bias, and systemic inequalities if not carefully governed.

THANK YOU!